The Belt and Road Initiative as a Catalyst for AI-based Agri-Engineering Applications of UAV in China, Japan, Bangladesh, and Tunisia

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The Belt and Road Initiative (BRI) has emerged as a transformative platform for advancing AI-driven precision agriculture through cross-border technological collaboration. This research proposal examines the integration of unmanned aerial remote sensing (UAS) into agricultural engineering frameworks under BRI, focusing on China's technological leadership, Japan's precision engineering expertise, and the localized applications in Bangladesh and Tunisia. By analyzing case studies in crop monitoring, irrigation optimization, and pest management, we demonstrate how UAS-enhanced AI systems address critical agricultural challenges while fostering sustainable development. The study highlights the role of BRI in bridging technological gaps, promoting knowledge transfer, and establishing standardized protocols for data-driven farming practices.

Key Words : *The Belt and Road Initiative; unmanned aerial remote sensing; precision agriculture*

1. INTRODUCTION

Since its launch, the Belt and Road Initiative (BRI) has expanded beyond infrastructure development into a multidisciplinary framework that now includes precision agriculture and AI innovation[1)-5)]. A key enabler of this shift: Unmanned Aerial Vehicles (UAVs)[6)-10)], equipped with multi-spectral cameras [11)-15)] and AI algorithms[15)-20)], which have become vital tools for precision agriculture − facilitating real-time crop monitoring, resource optimization, and climate-resilient farming.

Under the background of BRI, China, Japan (non-signed, but positive to it), Bangladesh, and Tunisia are collaboratively or individually applying Unmanned Aircraft Systems (UAS) technology for sustainable agricultural development, addressing shared challenges like food security, water scarcity, and labor shortages. Complementing this, the BRI Digital Silk Road[21)-30)] launched in 2013 to prioritize digital connectivity has opened unprecedented opportunities for cooperation between historically competitive nations, particularly in artificial intelligence. Tunisia, with its strategic North African position and ongoing digital transformation, serves as an ideal test-bed for such partnerships: On the day of the demonstration at the Higher School of Agricultural Engineering of Majaz al Bab in Tunisia, engineers from China's Unistrong Technology equipped a standard tractor with an autonomous driving system using China's BeiDou satellite navigation, successfully transforming it into the country's first unmanned agricultural vehicle capable of centimeter-level precision, as evidenced by its ability to repeatedly graze past stones placed in its path at varying speeds; Tunisian experts praised the technology's accuracy and potential for widespread application in agriculture across Arab and African regions, with the tractor set to be featured again at the upcoming China-Arab States BDS/GNSS Cooperation Forum in Tunisia[31)].

Fig.1 Precision Agriculture Illustrations of China, Japan, Bangladesh and Tunisia generated by DouBao AI.

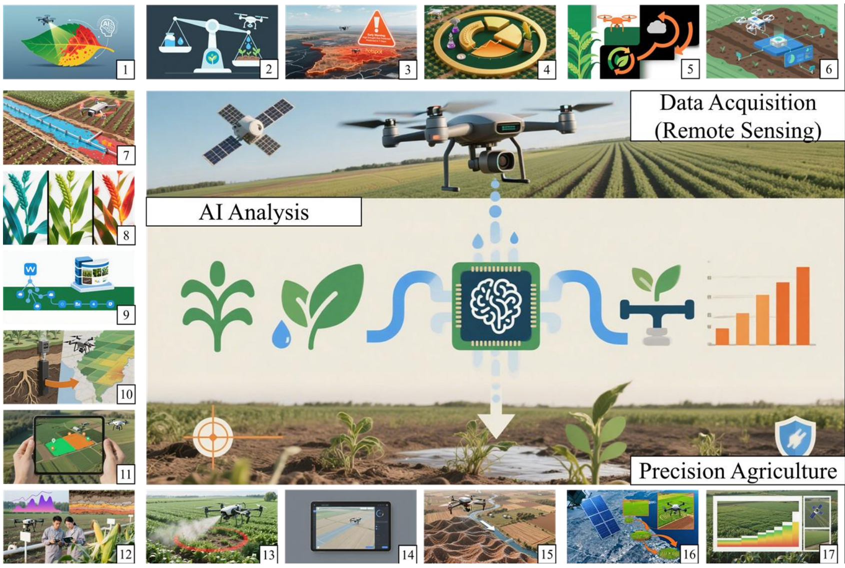

Fig.2 Illustrations of main tasks in the applications of Unmanned Aerial Remote Sensing (UAS) in Precision Agriculture, prompts and images are generated by YuanBao and DouBao AI, individually. Detailed information are included in Supplementary Material.

In Bangladesh, Commerce Adviser Sheikh Bashiruddin, leading a Bangladeshi delegation, formally requested assistance in drone technology for agriculture from Chinese Commerce Minister Wang Wentao during a meeting in Dhaka, highlighting its potential to revolutionize fertilizer application, sowing, pesticide spraying, and crop monitoring; in response, Minister Wang expressed China's readiness to collaborate in smart agriculture and drone technology to boost productivity, which culminated in the signing of two Memorandums of Understanding (MoUs) on establishing a Joint Working Group for free trade and e-commerce cooperation to strengthen bilateral trade and technological ties[32]. Japan, while not a formal BRI signatory, has aligned its "Free and Open Indo-Pacific" strategy, which is similar[33].

For China, Japan, Bangladesh, and Tunisia, the BRI's fusion of UAV technology and AI is more than a technical upgrade–it's a re-imagining of agricultural cooperation. Where once BRI focused on building roads and ports, it now builds resilience: by equipping farmers with tools to work smarter, not harder, and by connecting nations to solve problems that transcend borders. As climate change intensifies and global food demand rises, this shift could not come at a more critical moment–and the BRI’s role as a catalyst for AI-based precision agriculture may well define its legacy in the 21st century.

In this research, we also prefer to make a plan of collaborating these 4 countries together in the BRI, that they can apply the individual ability to contribute to the precision agriculture. As shown in Figure 1, an illustration was made for describing the on-site illustrations of using UAV, AI in the practical situations. Chinese UAV and AI devices can help the researchers to manage the agriculture field or water precisely. And Japanese agriculture management experience can also be good reference for the other countries. And Tunisia and Bangladesh are meaningful test-beds for these technologies and experience.

Furthermore, as shown in Figure 2, an illustration was made for describing the practical UAS application cases in precision agriculture. There are 3 main factors included in the workflow, i.e., Data acquisition by remote sensing approach (satellite and UAV, and UAV is the main topic in this research), AI-based data analysis and precision agriculture applications. Based on these application cases, the authors are trying to include them all in the BRI (some cases are developed already), and also attempt to design a reasonable framework for the sustainable developing routine that each country from this collaboration can benefit it.

2. TECHNOLOGICAL SYNERGY IN BRI-AI PARTNERSHIPS

(1) China Leads in AI and UAS Development

China has established itself as a global leader in artificial intelligence (AI) and UAS, providing the technological backbone for precision agriculture under BRI[34]. Through initiatives like the Digital Silk Road, China promotes the integration of AI, cloud computing, and IoT with agricultural practices[35]. For instance, the Institute of Remote Sensing and Digital Earth under the Chinese Academy of Sciences has developed advanced remote sensing technologies for real-time crop monitoring and drought prediction, which are being tailored to the needs of BRI partner countries[36]. Additionally, Chinese firms such as Huawei have built cloud computing infrastructure in Tunisia, enabling AI-driven data processing for agricultural applications[37]. China’s export of multi-purpose UAVs (e.g., Wing Loong and Rainbow series) to Arab states also demonstrates its capacity to scale technology adaptively[38]. These efforts are complemented by training programs, such as the Huawei ICT Academy, which has equipped thousands of students in BRI countries with skills in AI and 5G technologies[39].

(2) Japan Contributes to Precision Engineering

Though not a formal BRI signatory, Japan aligns its expertise in precision engineering and sustainable agriculture with BRI objectives through pragmatic cooperation frameworks such as the "Japan-China Third-Party Market Cooperation" model[40]. Japanese firms specialize in miniaturized UAV sensors and AI-based yield prediction software, which enhance resource efficiency in farming[41]. For example, collaborations under the Official Development Assistance (ODA) framework, such as the SATREPS project "Supporting System for Agricultural Water Management using Remote Sensing in Tunisia," have demonstrated the optimization of water usage for olive growers through UAV-based canopy analysis and other remote sensing technologies[42]. But it should be attentive, Japan is promoting its own "Free and Open Indo-Pacific" strategy, and while emphasizing the "similarity" with BRI, Japan is still not a BRI participating country. Japan’s strengths in standardized agricultural management and water-saving technologies – such as underground irrigation systems and climate-resilient crop varieties – provide valuable models for countries like Bangladesh and Tunisia, where fragmented landholdings and water scarcity pose challenges[43]. By sharing knowledge through multilateral platforms, Japan helps bridge technological gaps while promoting eco-friendly practices[44].

(3) Bangladesh and Tunisia Localized Applications

As testing grounds for UAV and AI innovations, Bangladesh and Tunisia focus on adapting technology to local conditions. In Bangladesh, UAVs equipped with hyperspectral imaging map soil moisture levels, enabling targeted irrigation and reducing water waste–a critical intervention in flood- and drought-prone regions[45]. Similarly, Tunisia leverages Chinese-funded AI projects to combat pests like wheat rust, with UAV swarms identifying outbreaks before they cause significant damage[46]. Both countries benefit from capacity-building initiatives; for instance, vocational schools in Tunisia integrate UAS maintenance into curricula under bilateral cooperation programs[47], while Bangladeshi farmers receive smartphone-based training to operate UAVs through initiatives supported by Department of Agricultural Extension (DAE)[48]. These localized approaches ensure that technology transfer translates into sustainable gains for smallholder farmers.

3. POSSIBLE CASE STUDY PLANS

(1) Precision Fertilization in Bangladesh

Bangladesh's intensive rice-farming system faces the dual challenge of ensuring food security for a growing population and mitigating the environmental degradation caused by the overuse of chemical fertilizers. Traditional blanket application methods lead to nutrient runoff, soil acidification, and high production costs.

The pilot project established an integrated "scan-and-apply" system. Chinese-developed multi-rotor UAVs conducted initial high-resolution surveys of the pilot areas, equipped with hyperspectral imaging sensors. This data was processed using cloud-based algorithms to generate detailed Digital Nutrient Maps that visualized variability in key soil nutrients like Nitrogen (N), Phosphorus (P), and Potassium (K) across individual fields.

These maps were then loaded into the variable-rate application (VRA) systems of fertilizer spreaders. Crucially, the system was dynamic. The project incorporated a network of Japanese-developed, low-cost wireless NPK sensors installed at various depths in the soil. These sensors provided real-time nitrogen level data, which was fed back into the system. The AI model could then make micro-adjustments to the fertilization prescription, accounting for daily changes in soil conditions and plant uptake.

The reduction in fertilizer use directly lowered input costs for farmers, while the yield increase significantly boosted their income. This improved the economic viability of smallholder farming. Reduced chemical runoff protected local water sources from eutrophication and decreased the carbon footprint of rice production. The success of the pilot has prompted discussions with the Bangladeshi government and agricultural banks to create financing models for custom hiring centers, where farmers can rent UAV and sensor services, making the technology accessible beyond large landholders.

(2) Drought Monitoring in Tunisia

Tunisia's agriculture, heavily reliant on olive oil and cereal production, is increasingly vulnerable to climate change-induced droughts and water scarcity. Inefficient irrigation practices were depleting critical aquifers.

The project moved beyond periodic assessment to a continuous monitoring system. A coordinated cluster of fixed-wing UAVs, capable of covering large swathes of land, was deployed on a regular flight schedule. These UAVs were equipped with multispectral and thermal cameras. The multispectral data provided detailed Normalized Difference Vegetation Index (NDVI) and Normalized Difference Water Index (NDWI) maps, indicating plant health and water stress long before it was visible to the naked eye. Thermal imaging identified variations in canopy temperature, a direct indicator of irrigation adequacy.

The raw aerial data was transmitted to a central platform developed in partnership with Chinese tech firms. Here, machine learning models fused the UAV data with satellite weather forecasts (predicting temperature, evaporation, and precipitation) and historical drought patterns. The output was not just a current snapshot but a predictive Drought Risk Index for different regions, forecasting water stress levels in advance.

The improvement in irrigation efficiency translated into massive water savings, allowing water authorities to better manage reservoir levels and aquifer recharge. Farmers received actionable alerts on a simple mobile application, advising them on the optimal timing and quantity for irrigation, moving from a schedule-based to a need-based approach. The accurate, data-driven drought models are now being used by the government to inform water allocation policies and provide targeted support to the most vulnerable farming communities.

(3) Pest Control in Japan-Bangladesh Rice Farms

Rice Blast Disease (Magnaporthe oryzae) is a devastating fungal pathogen capable of destroying entire rice crops. Preventative, blanket spraying of chemical pesticides is common but expensive, harmful to the ecosystem, and can lead to pesticide resistance.

This joint initiative created an early-warning AI-driven detection system. UAVs captured ultra-high-resolution RGB and multispectral imagery of rice fields at critical growth stages. The unique spectral signature of Rice Blast—slight changes in leaf reflectance that precede the formation of visible lesions—was the key.

Japanese institutions provided extensive image libraries of infected crops to train a convolutional neural network (CNN) . This AI algorithm was integrated into the data processing pipeline, where it could scan incoming UAV imagery and flag infected patches with high accuracy before widespread symptoms appeared. The system geo-tagged these hotspots.

Upon receiving early warnings via SMS, farmers could target the application of biopesticides or softer chemicals only to the identified hotspots, achieving a reduction in overall chemical usage. This preserved beneficial insects and reduced chemical residues in the grain. By preventing the spread of the disease, crop losses were minimized, securing both food supply and farmer livelihoods. The project served as a powerful capacity-building exercise. Bangladeshi agronomists and engineers received training from Japanese experts in both phytopathology and AI model management, building local expertise for sustainable pest management.

(4) Summary

Together, these case studies demonstrate a possible powerful global trend: the fusion of UAVs, IoT sensors, and AI is revolutionizing agriculture. The common thread is a shift from reactive, uniform practices to proactive, hyper-localized management. This precision agriculture approach is key to building more resilient, productive, and sustainable food systems worldwide, directly addressing challenges like resource scarcity, climate change, and environmental protection.

4. CHALLENGES AND MITIGATION STRATIGIES

The integration of advanced technologies like UAVs and AI in agriculture, while promising, faces significant hurdles that must be strategically addressed to ensure equitable and sustainable adoption.

(1) Technical and Regulatory Barriers

Beyond the initial proof-of-concept, widespread deployment faces obstacles such as the limited battery life of UAVs, which restricts operational time over large fields, and the variable quality of connectivity in rural areas needed for real-time data transmission. A critical, often overlooked barrier is data standardization; information collected by different hardware and software platforms can be incompatible, creating "data silos" that hinder comprehensive analysis. Furthermore, stringent regulatory frameworks in many countries classify agricultural UAVs under the same strict aviation rules as commercial drones, creating complex licensing and airspace permission processes that stifle innovation.

Successful projects, like the joint Japan-Bangladesh initiative on rice blast disease, demonstrate that collaboration is key. To overcome these barriers, a multi-stakeholder approach is essential. This includes developing open-source data protocols to ensure interoperability between systems from different manufacturers. Governments can play a pivotal role by creating "agri-tech sandboxes" − special economic zones with relaxed regulations for testing−and establishing clear, separate certification pathways for low-altitude agricultural UAVs. Partnering with local telecommunications companies to improve rural broadband infrastructure is also a crucial step to enable the cloud-based AI models that power these technologies.

(2) Capacity Building

The most advanced technology is futile if the end-users cannot operate or trust it. Smallholder farmers, who are the backbone of agriculture in many developing countries, often lack the digital literacy to interpret AI-generated data maps or maintain sophisticated equipment. There is also a natural resistance rooted in tradition and risk aversion; farmers are hesitant to adopt new practices without tangible, guaranteed proof of success from their peers. This creates a "know-do" gap where the existence of technology does not automatically translate to its effective use.

To bridge this gap, proactive and culturally sensitive training models are vital. Programs like China ’ s "demonstration villages" in Kenya are effective because they are led by local "champion farmers" who receive intensive training and then demonstrate the techniques to their community in a relatable, peer-to-peer manner. Beyond one-off workshops, establishing local technology service hubs creates sustainable expertise and employment. These hubs, often managed by local agronomists or entrepreneurs, can provide ongoing support, maintenance, and data interpretation services, building long-term trust and ensuring the technology is understood as a practical tool rather than a black-box solution.

(3) Financial Sustainability

The high initial capital expenditure for UAVs, sensors, and AI software is prohibitive for individual smallholder farmers. Even with proven returns on investment, the upfront cost remains a massive barrier. This financial challenge extends beyond the hardware to the "soft" infrastructure, including the costs of data storage, algorithm licensing, and technical support, creating a total cost of ownership that is often unaffordable.

Innovative financing and business models are required to overcome this hurdle. The strategy of leveraging large-scale investment funds, such as those associated with the Belt and Road Initiative (BRI), to subsidize initial rollout is a powerful catalyst. The most promising model is the

"UAV-as-a-Service" (UaaS) approach, exemplified in Pakistan. Here, BRI funding helps establish local service providers from whom farmers can lease UAV scanning and analysis services on a per-acre or subscription basis, paying from the increased revenue generated by the higher yields, thus eliminating the need for large capital outlays. This model not only makes the technology accessible but also stimulates local entrepreneurship and creates a self-sustaining market for agricultural technology services.

(5) Geopolitical and Partnership-Specific Challenges

While the general challenges of UAV/AI adoption are significant, this proposed four-country partnership (involving, for example, China, Japan, and two others) faces a unique set of complexities stemming from geopolitical tensions, differing regulatory regimes, and strategic competition. Failure to address these specifically could undermine the entire initiative.

The integration of Chinese-manufactured hardware (e.g., DJI drones) with Japanese-developed software and AI algorithms is a primary technical hurdle. This goes beyond simple API compatibility to deeper issues of data formats, communication protocols, and cybersecurity standards. Components and AI systems are often subject to strict export controls, especially when classified as "dual-use" (having both civilian and military applications). The partnership would need to navigate the complex web of regulations, which could restrict the sharing of critical technology between member states. Over-reliance on one partner's technology stack could create strategic dependencies, raising concerns about autonomy and leverage within the partnership.

The partnership would operate at the intersection of China's Cybersecurity Law and Personal Information Protection Law (PIPL) and other national data localization requirements. Where is the data collected by the UAVs stored? Who has access to it? How is sensitive information (e.g., imagery of critical infrastructure) handled? Establishing a legal framework for transferring data between jurisdictions with differing levels of data protection and state surveillance practices is a monumental challenge. Mutual trust is the cornerstone, yet it is often in short supply in the current geopolitical climate. A joint system becomes a shared target. A cyber-attack on the platform could lead to disputes over attribution and responsibility, potentially causing diplomatic friction between the partners.

Clear agreements are needed to distinguish between "background IP" (technology brought in by each partner) and "foreground IP" (new inventions resulting from the collaboration). How are the rights to foreground IP allocated? Is it joint ownership? If so, under which country's legal jurisdiction? How are IP infringements handled? What happens if a partner is accused of illegally transferring co-developed technology to a third party? The partnership requires a robust, mutually acceptable legal framework for dispute resolution that all parties trust, which may be difficult to establish. There is a risk of conflict if the partnership is perceived as one partner providing low-value hardware while another captures the high-value AI software and data analytic, leading to imbalances in the perceived benefits.

5. FUTURE DIRECTION

The successful pilot projects and ongoing mitigation efforts pave the way for more ambitious, integrated, and systemic applications of agricultural technology. The future of this field lies in moving from isolated successes to a globally connected, intelligent agricultural ecosystem.

(1) Standardization Initiatives

The current agri-tech landscape is fragmented, with different manufacturers using proprietary data formats and communication protocols. This lack of interoperability hinders data sharing, comparative analysis, and the creation of seamless, multi-technology solutions. For instance, soil data from a Japanese sensor may not be easily integrated with a Chinese UAV's imagery on a European data analytics platform.

Here is a critical need to promote international, cross-border standards for UAV data collection, sensor calibration, and AI model interoperability. Building on China's leadership in developing ISO standards for UAV perception and obstacle avoidance systems, future efforts should focus on creating open-data protocols for agricultural parameters (e.g., standard formats for NDVI, soil moisture, and pest incidence data). This would allow an AI model trained on data from Bangladeshi rice fields to be more easily validated and adapted for use in Vietnam or Brazil, dramatically accelerating global learning.

Universal standards would lower barriers to entry for new companies, foster healthy competition, and create a "plug-and-play" environment where farmers can mix and match best-in-class technologies without being locked into a single vendor. This is analogous to the USB standard in computing, which ensured compatibility across countless devices.

(2) Cross-Border Innovation Hubs

True innovation occurs when advanced technology is meaningfully adapted to local contexts. While hardware and software can be imported, the agronomic knowledge required to solve region-specific challenges resides with local farmers and researchers.

Establish physical joint innovation hubs in key agricultural regions like Tunisia or Bangladesh. These hubs would function as living laboratories, designed to combine Chinese UAV hardware expertise, Japanese prowess in AI software and precision, and deep local agronomic knowledge. The model can mirror the successful Sino-Africa Joint Research Centre in Kenya, which focuses on adapting technology for dryland farming. These hubs would not only test technology but also co-design solutions, train the next generation of local agri-tech experts, and serve as a launchpad for commercializing locally relevant products.

Such hubs would shift the dynamic from technology transfer to technology co-creation. A relevant example would be a hub in Tunisia focusing on developing AI algorithms specifically trained to recognize early signs of water stress in olive trees, a critical local crop, using data collected by UAVs flown by Tunisian researchers. This ensures solutions are culturally appropriate, economically viable, and sustainable.

(3) Climate-Resilient Frameworks

The ultimate application of this technology is to inform larger-scale decisions for climate resilience and resource security. Precision data from farms can be aggregated to model and manage entire agricultural systems.

Integrate UAV and satellite-collected data into macro-level climate-resilient frameworks. This includes using UAVs for high-resolution carbon sequestration monitoring in agricultural soils, which could unlock carbon credits for farmers practicing sustainable agriculture. Furthermore, virtual water trade analysis–using satellite data to assess the water footprint of crops–can optimize international food trade. For instance, satellite-assisted analysis could inform policy on virtual water trade between China and South Asian countries, encouraging the production of water-intensive crops in regions with greater water abundance and promoting water-efficient crops in arid regions.

This moves the focus from optimizing a single field to optimizing a regional or global food system. It provides policymakers with data-driven insights to make decisions that enhance collective food security while reducing the overall water and carbon footprint of agriculture. This strategic approach ensures that gains in individual farm productivity contribute meaningfully to global environmental sustainability goals.

6. DISCUSSION

The collaborative efforts under the Belt and Road Initiative (BRI) demonstrate a transformative shift from traditional infrastructure projects to a sophisticated, technology-driven partnership model for tackling global agricultural challenges. The cases of China, Japan, Bangladesh, and Tunisia reveal a powerful synergy: China provides the scalable hardware and digital infrastructure, Japan contributes precision engineering and management expertise, while Bangladesh and Tunisia serve as vital test-beds for localized application and innovation. This model, facilitated by the Digital Silk Road, successfully leverages the comparative advantages of each nation, as evidenced by the tangible outcomes in precision fertilization, drought monitoring, and pest control. However, the path to widespread adoption is not without obstacles; technical fragmentation, capacity gaps, and financial barriers present significant hurdles. The proposed mitigation strategies–such as standardizing data protocols, establishing innovation hubs, and promoting ‘UAV-as-a-Service’ models –are therefore critical for ensuring these technologies are accessible, sustainable, and ultimately owned by the local communities they are designed to serve. In conclusion, the integration of UAV and AI technologies under the BRI framework represents more than a mere technical upgrade; it signifies a fundamental re-imagining of international cooperation for food security. By fostering a collaborative ecosystem where technology, knowledge, and resources are shared, this initiative lays the groundwork for a more resilient and productive global agricultural system, capable of meeting the dual challenges of climate change and rising food demand. The success of this partnership model offers a replicable blueprint for other regions, positioning the BRI as a pivotal catalyst for sustainable development in the 21st century.

7. CONCLUSION

The synergy between China`s technological leadership, Japan’s precision engineering, and the localized adaptation in Bangladesh and Tunisia exemplifies how BRI can transform agriculture through innovation. By addressing challenges via coordinated policy, training, and financing, these partnerships can scale climate-resilient practices across the Global South.

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